

A VMO Morse Formula and a Complete Boundary-Extension Criterion for Unoriented Line Fields on Surfaces

Candidate solution packet for arXiv:1407.1707

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Status. Candidate full solution, likely valid, awaiting expert review. The theorem settles the proposed extension in the compact oriented surface/Q-tensor setting in which the source raises it.

Abstract

Canevari, Segatti, and Veneroni prove a Morse index formula for VMO vector fields on manifolds with boundary and propose extending it to VMO line fields. For a compact oriented surface N , the Q-tensors representing unoriented lines form the oriented plane bundle $\mathcal{E} = \text{Sym}_0^2(TN)$. Its transition functions have twice the angle of those of TN , so $e(\mathcal{E}) = 2e(TN)$. We use this observation to define an intrinsic half-integer line index and a boundary winding index for VMO data. We prove the exact formula

$$\text{ind}_{\text{line}}(Q, N) + \text{ind}_{\partial}^{\text{line}}(g) = \chi(N),$$

and show that a bounded, uniformly positive VMO boundary line field g has a bounded, uniformly positive VMO extension if and only if $\text{ind}_{\partial}^{\text{line}}(g) = \chi(N)$. Thus the same invariant gives both the Morse formula and a complete prescribed-boundary characterization.

1 Source question and precise scope

The source is G. Canevari, A. Segatti, and M. Veneroni, *Morse's index formula in VMO for compact manifolds with boundary*, arXiv:1407.1707 [1]. At the start of its Q-tensor application in Section 4 (source PDF page 17), after saying that it restricts to surfaces without boundary, it states:

“Clearly, it would be interesting to extend Theorem [the VMO Morse theorem] to VMO line fields.”

The complete source passage is reproduced in Figure 1.

The surrounding section works with a compact connected oriented surface $N \subset \mathbb{R}^3$ and identifies a line field with a uniformly nonzero section of its Q-tensor bundle. We solve exactly the corresponding problem when N has boundary. We also include the closed-surface specialization. For higher-dimensional manifolds, a line field is a section of $\mathbb{P}(TN)$ with fibre $\mathbb{RP}^{n-1}$; several obstruction groups can occur, so no all-dimensional one-integer classification is claimed here.

The two explicit “Question” environments earlier in the source are not open targets: Propositions in the same paper answer them. The present packet addresses only the separate line-field proposal quoted above.

4. AN APPLICATION: Q -TENSOR FIELDS AND LINE FIELDS.

In the mathematical modelling of Liquid Crystals two different theories are eminent. In the Frank-Oseen theory the molecules are represented by the unit vector field n which appears in the energy (1.1). The main drawback of this approach is to neglect the natural head-to-tail symmetry of the crystals. The theory of Landau-de Gennes takes this symmetry into account by introducing a tensor-valued field, called Q -*tensor*, to which is associated a scalar parameter s that represents the local average ordered/disordered state of the molecules. In the particular, but physically relevant, case when the order parameter is a positive constant, there is a bijection between Q -tensors and line fields. The differences between the vector-based and the line field-based theory have been studied in [1], in two- and three-dimensional Euclidean domains. In this Section we have two aims: firstly we apply some of the results obtained in Section 3 to line fields on a compact surface, obtaining the VMO-analogue of Poincaré-Kneiser Theorem (see Theorem 4.2 below). In particular, we restrict to the case of a surface without boundary. Clearly, it would be interesting to extend Theorem 1.4 to VMO line fields. Moreover, we show how the question of orienting a line field, studied in [1], has generally a negative answer on a compact surface. As it happens for liquid crystals in Euclidean domains, the elastic part of the Landau-de Gennes energy for nematic shells is, at least in some simplified situations, proportional to a Dirichlet type energy. See on this regard [17] and [23]. Therefore, owing to the embedding of Sobolev spaces in VMO spaces, (1.3), Proposition 4.3 establishes a relation between the existence of finite energy Q -tensors with strictly positive order parameter and the topology of the underlying surface, thus extending our application scope from the Frank-Oseen theory to the (constrained) Landau-de Gennes one, for uniaxial nematic shells.

4.1. Q -tensors and line fields. Nematic shells are the datum of a compact, connected and

Figure 1: The proposal in Section 4 of arXiv:1407.1707.

2 Definitions and the boundary invariant

Let N be a compact connected smooth oriented Riemannian surface. Write P_x for the orthogonal projection onto $T_x N$. Intrinsically, set

$$\mathcal{E} := \text{Sym}_0^2(TN),$$

the bundle of trace-free symmetric endomorphisms of TN . If $N \subset \mathbb{R}^3$ is oriented with unit normal γ , this is precisely the source's bundle

$$\mathcal{Q}_x = \{Q \in \text{Sym}_0(3) : Q\gamma(x) = 0\}.$$

For a unit tangent vector n , define

$$q(n) := \sqrt{2} \left(n \otimes n - \frac{1}{2} P \right). \quad (1)$$

Then $|q(n)| = 1$ and $q(n) = q(-n)$. Conversely, a unit element of $\mathcal{E}_x$ has a unique positive eigenspace, so the unit-circle bundle $S(\mathcal{E})$ is canonically the bundle of unoriented tangent lines.

Assume for now that $\partial N \neq \emptyset$. Let ν be the outward unit normal to ∂N inside TN , and orient each boundary component by the unit tangent τ for which (ν, τ) is a positive frame of TN . The canonical boundary line section is

$$q_\tau := q(\tau) \in S(\mathcal{E}|_{\partial N}). \quad (2)$$

If g is a continuous nonzero section of $\mathcal{E}|_{\partial N}$, normalize it to $\widehat{g} = g/|g|$. The orientation and metric of $\mathcal{E}$ give a fibrewise quarter-turn $J_{\mathcal{E}}$. There is a unique map $r_g : \partial N \rightarrow S^1$ such that

$$\widehat{g} = (\Re r_g)q_{\tau} + (\Im r_g)J_{\mathcal{E}}q_{\tau}. \quad (3)$$

Define the total relative winding, with the induced orientations on all boundary components, by

$$w(g) := \sum_{C \subset \partial N} \deg(r_g|_C) \in \mathbb{Z}. \quad (4)$$

For VMO data, we use the same averaging-and-fibre-projection convention as the source. Namely, embed $\mathcal{E}$ smoothly in a trivial Euclidean bundle, take local averages of g at scale ε , and project the result orthogonally to $\mathcal{E}_x$. Denote the resulting continuous section by g_{ε} . If

$$g \in \text{VMO}(\partial N, \mathcal{E}), \quad 0 < c \leq |g| \leq C < \infty \quad \text{a.e.}, \quad (5)$$

then g_{ε} is nonzero for all sufficiently small ε , and $w(g_{\varepsilon})$ is independent of that small scale. We therefore set

$$w(g) := w(g_{\varepsilon}). \quad (6)$$

The bundle-VMO lemma below proves that this is well-defined. We likewise write $e(\mathcal{E}, g)$ for the stable relative Euler class $e(\mathcal{E}, g_{\varepsilon})$ when g is VMO.

Definition 1 (Line and boundary indices). Let $Q \in \text{VMO}(N, \mathcal{E})$ have VMO trace g satisfying (5). Let Q_{ε} be a source-type continuous regularization, nonzero on ∂N . Its ordinary plane-bundle index is the sum of the integer local degrees of a transverse perturbation, and is denoted $\text{Ind}_{\mathcal{E}}(Q, N)$. Define

$$\text{ind}_{\text{line}}(Q, N) := \frac{1}{2} \text{Ind}_{\mathcal{E}}(Q, N), \quad \text{ind}_{\partial}^{\text{line}}(g) := -\frac{1}{2} w(g). \quad (7)$$

Both may be half-integers. This is the standard normalization in which a half-strength nematic defect has line index $1/2$.

3 Idea of the proof

The sign ambiguity that blocks a literal “points inward” definition is removed by representing an unoriented line as the Q-tensor $q(n)$. In an oriented tangent frame, rotating n by θ rotates $q(n)$ by 2θ . Thus the Q-bundle has twice the Euler class of TN .

Along the boundary, the tangent line gives a canonical reference section q_{τ} . Relative to it, the boundary datum has an integer winding $w(g)$. The relative Euler number of the Q-bundle is exactly

$$2\chi(N) + w(g).$$

It is also the total ordinary index of any Q-section with boundary value g . Dividing by two and moving the winding term to the left gives the desired Morse formula.

The analytic VMO step introduces no new topological obstruction. The source’s local-average projection works for any smooth Euclidean subbundle, not only TN , so it applies to $\mathcal{E}$. Uniform positivity prevents the regularized Q-fields from acquiring zeros. Conversely, if the relative Euler number vanishes, average the VMO datum into a collar and use the vanishing Euler obstruction to extend the resulting continuous nonzero section over the two-dimensional core. The glued field is VMO, has the prescribed trace, and can be radially clamped to uniform positive upper and lower bounds.

4 Topological calculation

Lemma 2 (The doubled Euler bundle). *The bundle $\mathcal{E} = \text{Sym}_0^2(TN)$ is an oriented metric plane bundle. With the orientations above,*

$$\langle e(\mathcal{E}, q_\tau), [N, \partial N] \rangle = 2\chi(N). \quad (8)$$

For any continuous nonzero boundary section g ,

$$\langle e(\mathcal{E}, g), [N, \partial N] \rangle = 2\chi(N) + w(g). \quad (9)$$

Proof. In a positively oriented orthonormal tangent frame (e_1, e_2) , use the orthonormal frame

$$A_1 = \frac{e_1 \otimes e_1 - e_2 \otimes e_2}{\sqrt{2}}, \quad A_2 = \frac{e_1 \otimes e_2 + e_2 \otimes e_1}{\sqrt{2}} \quad (10)$$

of $\mathcal{E}$. If (e_1, e_2) is rotated through θ , direct expansion shows that (A_1, A_2) is rotated through 2θ . Thus the oriented frame transition maps of $\mathcal{E}$ are the squares of those of TN .

The relative Euler number of (TN, τ) is $\chi(N)$. One way to see the sign is to apply the continuous Morse formula from [1] to a tangent vector field whose boundary value is τ : its inward boundary set is empty, so its total zero index is $\chi(N)$. Squaring every transition map and the boundary clutching map doubles the obstruction cocycle. Since $q(\tau)$ is precisely the squared boundary section, this proves (8).

Changing a nonzero boundary section from q_τ to g changes the relative Euler obstruction by the total degree of the transition map r_g in (3). With the induced boundary orientation, the sign is positive. For example, on the positively oriented disk q_τ has degree 2 in a constant trivialization, while a constant section has relative winding -2 and relative Euler number 0. Hence

$$\langle e(\mathcal{E}, g), [N, \partial N] \rangle = \langle e(\mathcal{E}, q_\tau), [N, \partial N] \rangle + \deg(r_g),$$

which is (9). □

Lemma 3 (Relative index and extension). *Let $E \rightarrow M$ be an oriented metric plane bundle over a compact connected oriented surface with boundary, and let h be a nonzero continuous section of $E|_{\partial M}$.*

- (i) *The total local index of any continuous extension, after a transverse perturbation fixed near the boundary, is $\langle e(E, h), [M, \partial M] \rangle$.*
- (ii) *The section h extends to a continuous nowhere-zero section on M if and only if this integer is zero.*

Proof. Triangulate the pair $(M, \partial M)$ and normalize sections to the circle bundle $S(E)$. Since S^1 is path connected, h extends over the relative one-skeleton. The winding around each oriented two-cell is the obstruction to extending over that cell. These windings form a relative 2-cocycle with values in $\pi_1(S^1) = \mathbb{Z}$; its class is the relative Euler class. For a transverse section, the same cell degrees are the signed local zero indices, proving (i).

There are no higher cells and no higher obstructions. Thus a nonzero extension exists exactly when the relative obstruction class vanishes. By Poincaré–Lefschetz duality, $H^2(M, \partial M; \mathbb{Z}) \cong \mathbb{Z}$ for connected oriented M , and its evaluation on $[M, \partial M]$ detects the class. This proves (ii). □

5 The VMO passage

We isolate the precise analytic input. It is the source's construction with the tangent bundle replaced by a smooth Euclidean subbundle.

Lemma 4 (VMO bundle regularization and collar). *Let $E \rightarrow M$ be a smooth metric vector bundle embedded as a smooth subbundle of $M \times \mathbb{R}^m$.*

- (a) *A VMO section with VMO trace admits continuous local-average regularizations obtained by orthogonal projection to E_x . If the trace is bounded away from zero, the regularizations are uniformly nonzero on the boundary for small scales; their relative indices and boundary windings are then independent of the scale.*
- (b) *If $s \in \text{VMO}(M, E)$ satisfies $c \leq |s| \leq C$ almost everywhere, its regularizations satisfy $c/2 \leq |s_\varepsilon| \leq 2C$ everywhere for all sufficiently small ε .*
- (c) *If $h \in \text{VMO}(\partial M, E|_{\partial M})$ satisfies $c \leq |h| \leq C$, then on a sufficiently thin collar $\Phi : \partial M \times [0, r] \rightarrow M$ there is a VMO section H , continuous away from ∂M , with trace h , with $H|_{\Phi(\partial M \times \{r\})}$ continuous, and with $c/3 \leq |H| \leq 3C$.*

Proof. We indicate why the proofs in Sections 3–4 of [1] apply without a hidden tangent-vector assumption. In local coordinates, average the ambient $\mathbb{R}^m$ -valued representative and apply the orthogonal projection P_{E_x} . The map $x \mapsto P_{E_x}$ is smooth. The error made by replacing P_{E_y} by P_{E_x} on a ball of radius ε is therefore $o(1)$ uniformly; the remaining error is exactly the VMO mean oscillation. The trace construction on the double gives uniform compatibility between the interior and boundary regularizations. Homotopy invariance then makes the indices constant for small scale. These are precisely the ingredients used for the tangent bundle in the source.

For (b), on every sufficiently small ball B ,

$$|s_B| \geq \frac{1}{|B|} \int_B |s| - \frac{1}{|B|} \int_B |s - s_B| \geq c - o(1),$$

uniformly in the centre, and smooth fibre projection changes this by another $o(1)$. The upper bound is analogous.

For (c), put $x = \Phi(y, t)$ and take

$$H(x) = P_{E_x} \left(\frac{1}{|B_t^{\partial M}(y)|} \int_{B_t^{\partial M}(y)} h(z) \, dz \right), \quad t > 0.$$

The VMO oscillation estimate gives the stated bounds and continuity for $t > 0$. The standard VMO collar lemma proved in [1] says that this averaging extension has trace h and is VMO across $t = 0$; composition with the smooth family P_{E_x} preserves those properties. This proves (c). $\square$

6 Morse formula and full characterization

Theorem 5 (VMO Morse formula for line fields). *Let N be a compact connected oriented smooth surface with nonempty boundary, and put $\mathcal{E} = \text{Sym}_0^2(TN)$. Let $Q \in \text{VMO}(N, \mathcal{E})$ have VMO trace $g \in \text{VMO}(\partial N, \mathcal{E})$ satisfying $0 < c \leq |g| \leq C$. Then the indices in Definition 1 are well-defined and satisfy*

$$\text{ind}_{\text{line}}(Q, N) + \text{ind}_\partial^{\text{line}}(g) = \chi(N). \quad (11)$$

Equivalently,

$$\text{Ind}_\mathcal{E}(Q, N) = 2\chi(N) + w(g). \quad (12)$$

Proof. Lemma 4(a) gives a continuous regularization Q_ε that is nonzero on the boundary and has boundary value homotopic through nonzero sections to g_ε . By Lemma 3(i), its total integer index is

$$\text{Ind}_\mathcal{E}(Q, N) = \langle e(\mathcal{E}, g_\varepsilon), [N, \partial N] \rangle.$$

Lemma 2 and the definition of the stable VMO winding give

$$\text{Ind}_\mathcal{E}(Q, N) = 2\chi(N) + w(g).$$

Dividing by two and using (7) proves (11). $\square$

Theorem 6 (Prescribed-boundary characterization). *Under the hypotheses of Theorem 5, assume only that the boundary datum g is given. The following are equivalent:*

- (i) *There exists $Q \in \text{VMO}(N, \mathcal{E})$ with trace g and constants $0 < c' \leq C' < \infty$ such that*

$$c' \leq |Q(x)| \leq C' \quad \text{for almost every } x \in N.$$

Thus Q is a bounded, uniformly positive VMO line field.

- (ii) *The boundary line index satisfies*

$$\text{ind}_\partial^{\text{line}}(g) = \chi(N). \quad (13)$$

- (iii) *The explicit winding condition*

$$2\chi(N) + w(g) = 0 \quad (14)$$

holds.

Proof. The equivalence of (ii) and (iii) is the definition of the boundary index. If (i) holds, Lemma 4(b) makes all sufficiently small regularizations nonzero on the whole of N . Hence $\text{Ind}_\mathcal{E}(Q, N) = 0$, and Theorem 5 gives (ii).

Assume (iii). Apply Lemma 4(c) to extend g as a uniformly nonzero VMO section H over a collar U_r . On the inner collar boundary C_r , the section $h = H|_{C_r}$ is continuous and nonzero. Let $N_r = N \setminus U_r$. Additivity of the relative Euler obstruction under gluing and the absence of zeros in the collar give

$$\langle e(\mathcal{E}|_{N_r}, h), [N_r, C_r] \rangle = \langle e(\mathcal{E}, g), [N, \partial N] \rangle = 2\chi(N) + w(g) = 0.$$

By Lemma 3(ii), h extends over N_r to a continuous nowhere-zero section K . Glue K to H . The result is VMO, has trace g , and is uniformly positive and bounded because N_r is compact and both pieces are nonzero.

If desired, compose fibrewise with a Lipschitz radial clamping map that is the identity on the essential range of g . This preserves VMO and the trace and gives any convenient finite positive bounds. Thus (i) holds. $\square$

Corollary 7 (Closed surfaces). *If N is compact, connected, oriented, and has no boundary, every VMO Q -field for which the line index is defined satisfies*

$$\text{ind}_{\text{line}}(Q, N) = \chi(N).$$

In particular, a bounded uniformly positive VMO line field exists if and only if $\chi(N) = 0$.

Proof. The ordinary total index of a section of $\mathcal{E}$ is $\langle e(\mathcal{E}), [N] \rangle = 2\chi(N)$ by the double-angle calculation in Lemma 2. Divide by two. Necessity for a nonzero VMO line field follows from Lemma 4(b), and sufficiency when $\chi(N) = 0$ is the closed-surface rank-two obstruction statement (or the source's continuous Poincaré–Kneiser theorem, whose continuous section is VMO). $\square$

7 Checks, examples, and limitations

Disk sign check. Trivialize TN on the unit disk by the Cartesian frame. Along the positively oriented circle, $\tau(\theta) = (-\sin \theta, \cos \theta)$, so q_τ has phase $2\theta + \pi$ and degree 2. A constant boundary Q-field has relative winding -2 and extends constantly. Formula (9) gives $2\chi(D^2) - 2 = 0$, as required.

Annulus check. The induced orientations on the two boundary circles are opposite, and $\chi = 0$. The contributions of q_τ cancel. A constant Q-field has $w = 0$ and extends, again agreeing with the criterion.

Gauge invariance. Changing an auxiliary tangent frame by angle α changes Q-coordinates by 2α for both g and q_τ . Their relative phase, hence $w(g)$, is unchanged.

Scope. The result is complete for the compact oriented surface/Q-tensor problem posed by the surrounding source section. It does not classify line fields on nonorientable surfaces or arbitrary higher-dimensional manifolds. The latter is generally an obstruction-theory problem for $\mathbb{P}(TN)$ rather than a single Euler-number formula.

Novelty check. Exact-phrase and close-variant web/arXiv searches, the run indexes, and the complete OpenAlex citing-work list for the published source were checked on 11 August 2026. No later explicit solution of this VMO boundary problem was found. The continuous relative-Euler ingredients are classical; the novelty claim concerns their exact line-field normalization and the VMO trace/extension theorem. See `novelty_search.md` for the search bounds.

Computational sanity check. The included script `code/verify_transition_and_winding.py` verifies the double-angle matrix identity and the disk/annulus winding signs on a dense mesh. These computations are checks, not part of the proof.

8 Human-review recommendation

This packet should receive expert review as a full candidate. The most useful checks are:

1. verify the sign in $\langle e(\mathcal{E}, g), [N, \partial N] \rangle = 2\chi(N) + w(g)$ under the stated boundary orientation;
2. confirm that the source’s double/trace and collar arguments depend only on a smooth Euclidean subbundle and therefore apply to $\mathcal{E}$;
3. verify the gluing identification of the relative Euler class on the collar core;
4. compare the term “extend Theorem to VMO line fields” with the source authors’ intended defect-index normalization.

References

- [1] G. Canevari, A. Segatti, and M. Veneroni, *Morse’s index formula in VMO for compact manifolds with boundary*, Journal of Functional Analysis 269 (2015), 3043–3082; arXiv:1407.1707.